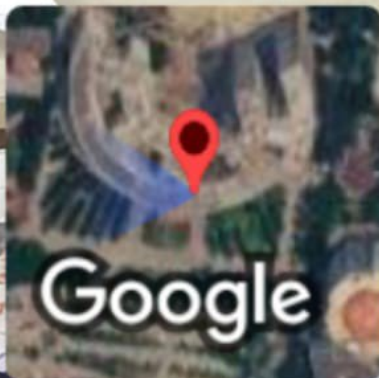


Presentation

DSS- Decision Support System

GPS Map Camera



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Lat 13.025835° Long 80.01662°

Wednesday, 15/07/2026 11:52 AM GMT +05:30

Query processing in Business Intelligence (BI) refers to the end-to-end mechanism by which a user's analytical question—whether submitted via a drag-and-drop dashboard, an ad-hoc filter, or a natural language prompt—is translated, optimized, executed against a data store, and returned as a visual insight. [Databricks](#)

Unlike traditional transactional databases (OLTP) built for high-speed row insertions, BI query processing is optimized for **Online Analytical Processing (OLAP)**—handling massive read-heavy aggregations across historical data. [MongoDB](#)

The BI Query Processing Lifecycle

When an end-user interacts with a BI dashboard or types a search query, the system typically drives through the following stages:

[User Interaction]

↓ (Clicks filter, changes

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[Data Source Execution]



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1. User Interaction & Abstraction

- **The Action:** The user interacts with the BI frontend (e.g., filtering a sales dashboard by region, drilling down into product categories, or typing *"Show me monthly revenue for Q3"* into a conversational analytics tool).
- **The Semantic Layer:** Modern BI tools use a semantic model (like Looker's LookML or Power BI's tabular models) to map abstract business terms ("Revenue", "Active Users") to underlying database schema objects (tables, columns, foreign keys) without exposing users to raw database architecture.

2. Translation & Query Generation

- Once the user defines the parameters, the BI tool dynamically builds a structured query—typically SQL (Structured Query Language), MDX (Multidimensional Expressions), or a



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- For ad-hoc queries, the system



3. Query Optimization

Query performance is critical in BI because datasets span gigabytes to petabytes. Optimization happens at two levels:

- **BI-Level Caching & Aggregation**
Awareness: Before hitting the database, the BI tool checks if the exact result or a compatible aggregate is already cached locally or in a summary table. If an aggregate table exists that already pre-computes monthly totals, the tool reroutes the query to scan that smaller table instead of raw transaction logs.
- **Database/Data Warehouse Optimizer:**
Once the query reaches platforms like Snowflake, BigQuery, or Databricks, the database query optimizer evaluates execution plans. It determines the optimal join strategy (e.g., hash joins vs. merge joins), applies predicate pushdown (filtering data *before* joining tables), and leverages columnar storage layouts.  Had only the columns requested.



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5. Data Rendering

The database returns a raw dataset (rows and columns of aggregated numbers). The BI tool takes this response and maps it to the appropriate visual components—rendering it into bar charts, scatter plots, heatmaps, or KPI cards for the user.

Modern Trends in BI Query Processing

- **Natural Language to Query (NLQ):**
Tools like Databricks Genie, Microsoft Copilot in Power BI, and ThoughtSpot use Generative AI and Large Language Models (LLMs) to translate plain English questions directly into optimized SQL queries, drastically lowering the technical barrier for business users.
- **Semantic Caching with AI:** Modern data stacks use semantic caching, where the system understands the *intent* behind a query rather than just the exact syntax. If a user asks a variation of a recently run query, the system can serve cached insights



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